Decomposing Lahore Air Quality (2019–2024)

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Group Members

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Research Question

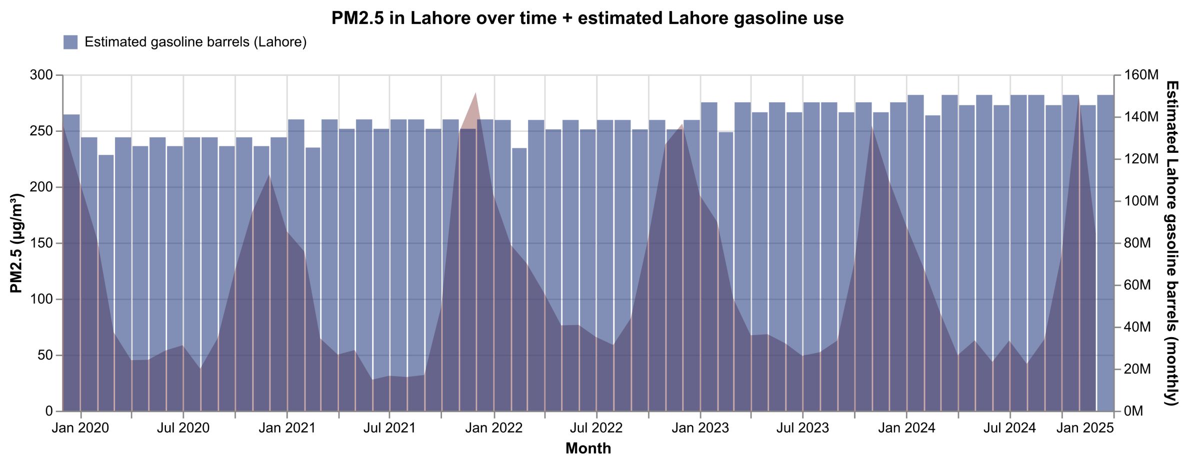
Lahore has experienced some of the highest air pollution levels recorded globally in recent years. Policymakers frequently attribute this problem to several competing explanations including agricultural burning, traffic emissions, and cross-border pollution. Our project evaluates these claims using multiple datasets to examine how Lahore's air quality has evolved over time and what factors may be associated with these changes. We want to see what factors are associated with the deterioration of air quality in Lahore, and do available data suggest that current urban policies are worsening conditions?

To address this question, we analyze three related dimensions: Air quality trends measured through PM2.5 monitoring stations, Urban development and activity using satellite night-time light intensity and Urban environmental change using satellite vegetation indices (NDVI).

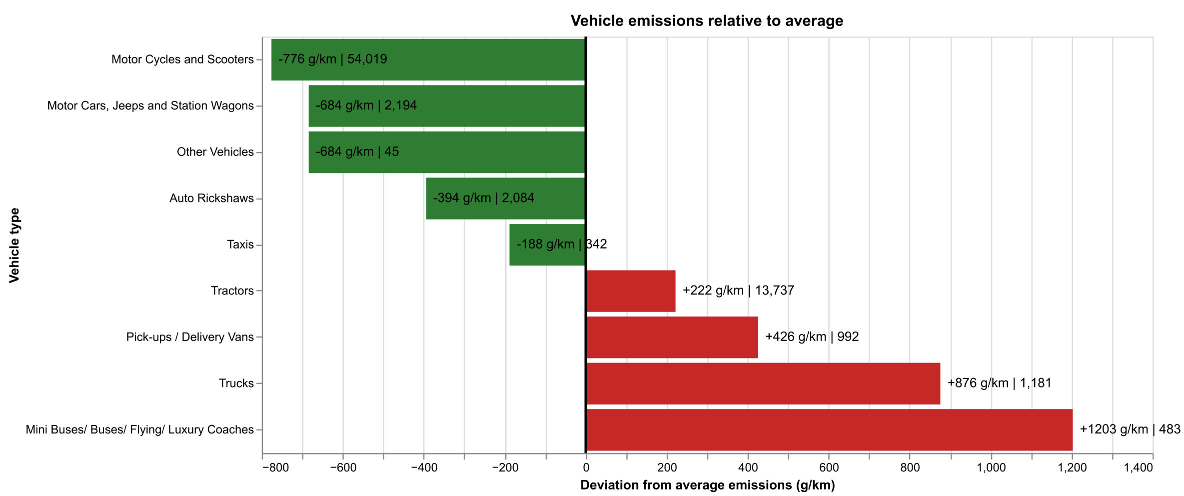
Data and Methodology

Our analysis integrates several datasets covering the period 2019–2024. We use hourly particulate matter measurements from the Pakistan Air Quality Initiative (PAQI). These measurements are aggregated into monthly averages for each sensor. Two satellite datasets were processed using Google Earth Engine. VIIRS Nighttime Lights which is used as a proxy for economic activity and urban densification NDVI (Normalized Difference Vegetation Index) which measures vegetation greenness and provides a proxy for urban green space. Both datasets were spatially averaged across a bounding box covering Lahore to produce monthly indicators. We incorporate the Punjab motor vehicle registration data and the Energy Institute global gasoline consumption data. Vehicle counts were aggregated by type (motorcycles, cars, trucks, buses) and combined with emission intensity estimates to approximate potential emission contributions. We then combine all indicators into a unified monthly panel dataset.

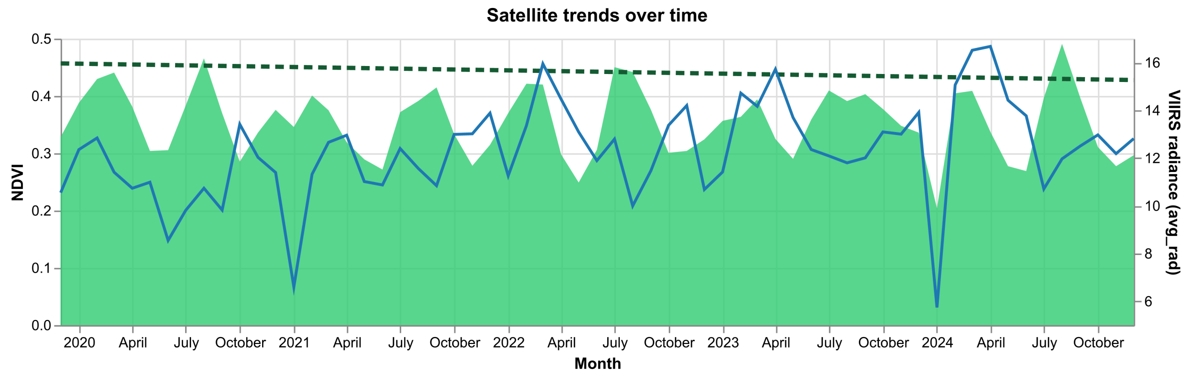
Static Visualizations and Findings



Key Finding: Despite relative stability in estimated fuel consumption across the period, PM2.5 levels show seasonal variation. This pattern suggests that fuel consumption alone does not explain air quality deterioration, pointing instead to seasonal meteorological factors or non-fuel pollution sources.



Key Finding: Motorcycles and scooters dominate vehicle registrations yet emit significantly below-average emissions per kilometer. Buses, trucks, and heavy vehicles emit 2–8 times the fleet average. This suggests that while heavy-duty vehicles are the primary emitters per-unit. Policy interventions targeting heavy vehicle traffic and fuel quality may be more cost-effective than restrictions on two-wheelers.



Key Finding: The NDVI shows an overall reduction as compared to the average (dashed line), reflecting vegetation removal. Nighttime light (VIIRS) intensity remains relatively stable with modest increases, suggesting moderate economic expansion in traffic. The decoupling between nightlights and PM2.5 suggests that economic growth alone does not drive pollution increases at this rate, but the removal of vegetation may be a contributor.

Streamlit Dashboard

We developed a Streamlit dashboard to demonstrate that air quality deterioration in Lahore is decoupled from fuel consumption trends, pointing to seasonal meteorological patterns and non-vehicular pollution sources as primary drivers. The static analysis presented above forms the analytical backbone of the dashboard. The dashboard includes:

1. Dual Interactive Maps: A side-by-side comparison of air quality (PAQI monitors) against satellite layers (VIIRS nightlights or NDVI), allowing users to visually inspect month-by-month co-variation.
2. Regional Vehicle Analysis: Dynamic filtering by region (Lahore, Sheikhupura) with real-time computation of total vehicles, weighted-average emissions, and aggregate emission profiles.
3. Satellite Imagery Layers: Monthly pre-rendered satellite PNG overlays (VIIRS and NDVI) from Google Earth Engine, displayed directly on folium maps with automatic color-scaling.
4. Real-time AQI Lookup: Integration with OpenWeatherMap API for live air quality queries at user-specified locations.